

INNOVATIVE MINING SYSTEMS

Mine Back Filling

By

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graph TD; A[Mine Back Filling] --> B[O/C mine]; A --> C[U/G coal mine]; A --> D[U/G metal mine]; B --> E[Dry filling]; C --> F["1. Hydraulic<br/>• Sand<br/>• Bottom ash<br/>• Processed O/B<br/>• Slag<br/>• Pond ash<br/>2. Dry filling"]; D --> G["• Cemented rock filling (CRF)<br/>• Hydraulic filling<br/>• Cemented hydraulic filling<br/>• High concentration filling<br/>• Cemented paste backfilling/Paste filling"];
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O/C mine

Dry filling

U/G coal mine

- 1. Hydraulic**
 - Sand
 - Bottom ash
 - Processed O/B
 - Slag
 - Pond ash
- 2. Dry filling**

U/G metal mine

- **Cemented rock filling (CRF)**
- **Hydraulic filling**
- **Cemented hydraulic filling**
- **High concentration filling**
- **Cemented paste backfilling/Paste filling**

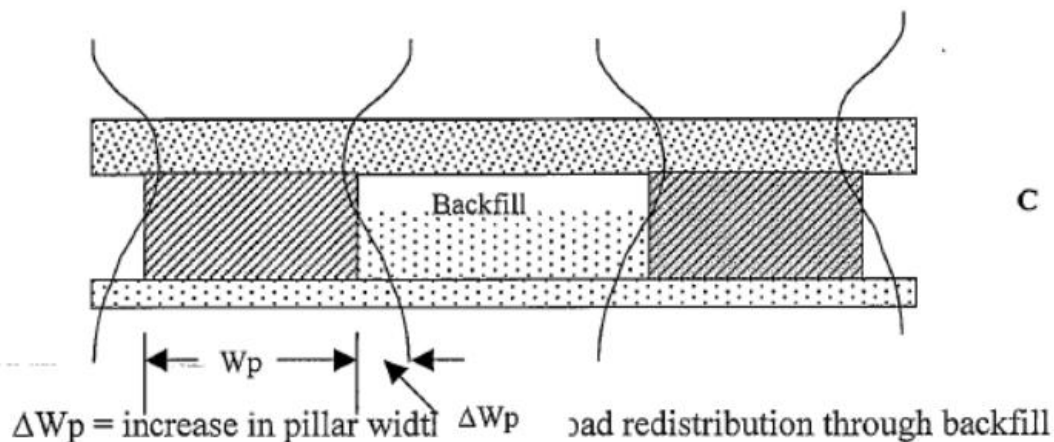
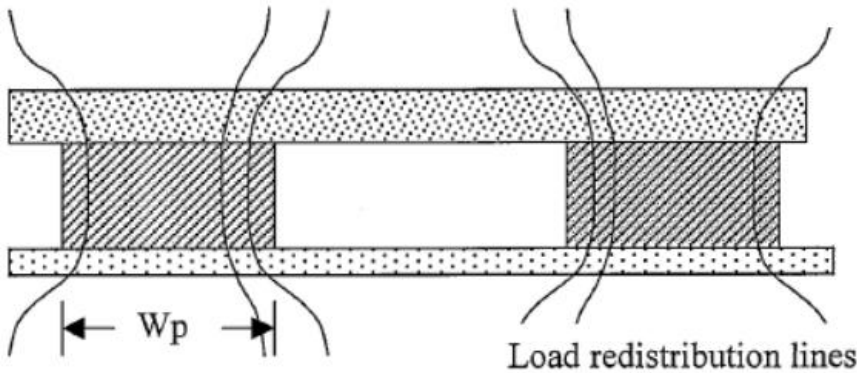
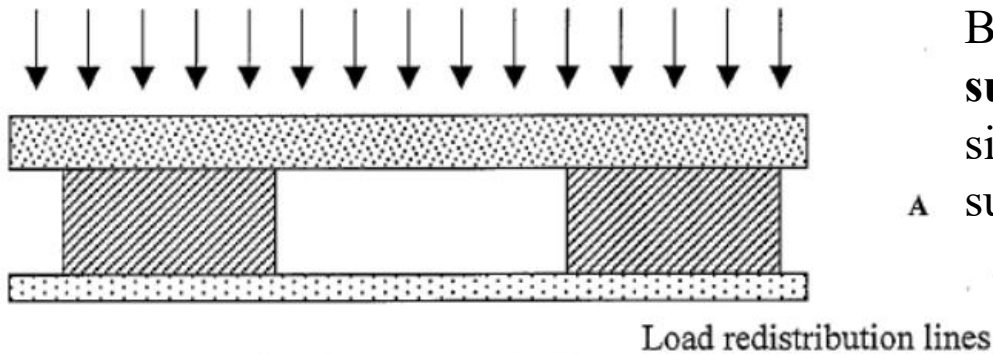
GROUND CONTROL MECHANISM OF BACKFILLING

Backfilling of underground mine workings after mineral extraction has been recognized primarily as one of the important techniques of ground control and is now employed universally in several coal and metalliferous mines not only in India but abroad as a routine work. However, to date there has been no well-established theory clearly expounding the supporting mechanism of the stowed mass or backfill (Zhang et al. 2018a).

It is generally considered that the backfill supports the rock stress and results into minimal bed separation and sagging of the rock into the opening, and as a result reduces the surface subsidence. It should be remembered that the mechanical properties and packing densities of the fill materials placed in the voids are much smaller than those of the rocks previously existing there before excavation and where fill is subsequently placed. Nevertheless, backfilling, even a partial filling method in coalmines, is widely adopted for ground stabilization (Gupta, 1996 and Behera et al. 2020b).

In metalliferous mines, the ground support function is the most important role for fill as a structural component (Mitchell et al. 1982; Chen and Jiao, 1991; Zou and Nadarajah, 2006; Dirige et al. 2009; Li and Aubertin, 2012; Behera et al. 2023). The presence of fill in any opening can prevent large scale movements and collapse of opening merely by occupying the void. If the fill is sufficiently stiff it can accept some part of the load previously carried by the ore and prevent the load being transferred into the remaining ore. Likewise, stability of the working face can be maintained by maximizing possible ore recovery. Fill acts as a support and can reduce subsidence of workings at higher levels and at the surface, or reduce the load on pillars. The fill as a ground support medium responds by deforming under the imposed back and wall loads.

Load redistribution



Backfilling is primarily a **passive support**, though in some mining situations it can also provide active support effects.

- Passive support acts after rock/ground movement occurs.
- Backfill does not apply a pre-load to the rock mass.
- It becomes effective only when the surrounding rock starts to deform and transfers load onto the backfill.

Aspect	Active support	Backfilling
Examples	Rock bolts, cable bolts, props	Sand fill, hydraulic fill, paste fill
Pre-loading	Yes	No
Acts before movement	Yes	No
Classification	Active	Passive

Brief history: UG coal stowing

- Past 125-130 years stowing is being practiced throughout the world.
- 1st hydraulic stowing -1864 , Pennsylvania, USA, Idea was given by Catholic Priest to prevent his church.
- The blessing of hydraulic stowing due to flood accidentally discovered in England (Mary Hey Pit).
- 1900-Other European countries.
- 1901-Polland, 1st introduced.
- 1914-France, 1st introduced.
- 1914-More than 100 mines in Germany.
- In India stowing in form of dry manual packing 1st tried in-Mohapani Colliery in Central India (1893)
- Hydraulic sand stowing was first started in Ballarpur Colliery (now in WCL) in March, 1913.
- 1915-Jogta Colliery, Jharia, 1916- Sanctoria Colliery (Raniganj), 1918- Seetalpur II pit, 1919- Bhowra (Jharia), contd..

Brief history: UG metal backfilling

- Mining with hydraulic backfills was introduced in Indian metalliferous mines for the first time in the year 1953 at Kolar Gold Field (KGF), Karnataka. At present around 20 hydraulic backfill plants and 5 paste fill plants are already in operations in metalliferous mines in India.
- The initial backfilling was started with sand and gradually backfilling was shifted to mill tailings. Further upgradation to the backfilling at KGF was done during 1985 (Choudhury and Ghose, 1991). At Mosabani group of mines in Singhbhum District of Bihar (presently in Jharkhand), mining with backfilling was practiced for extraction of chalcopryite ore bodies dipping at 25-35° (Shetigar, 1983). In continuation with the application of hydraulic backfilling in different hard rock mines, Jaduguda mines, UCIL had also implemented tailings based hydraulic backfilling during 1980s (Choudhury and Ghose 1991).
- In 1977, hydraulic backfilling using tailings was introduced to extract gold deposits at Hutti Gold Mines, Karnataka using overhand cut and fill mining (Thakur et al. 1992). Subsequently, hydraulic backfilling using cemented fill was applied to fill the Large Diameter Bore Hole (LDBH) stopes for maximizing the recovery from the rib pillars.
- Further, backfilling using sand was implemented in Rajpura Dariba mine, HZL, Rajasthan during 1985 and later sand was completely replaced by tailings (Rathore and Kumar, 1992). Post pillar stoping with backfilling was deployed for the first time in Mosabani Group of Copper mines in 1973 and it was also introduced in Rajpura-Dariba mines for extracting thick lead-zinc ore bodies using Vertical Retreat Mining (VRM) and Blast Hole (BH) stopes 1990 onwards (Choudhury and Ghose, 1991).

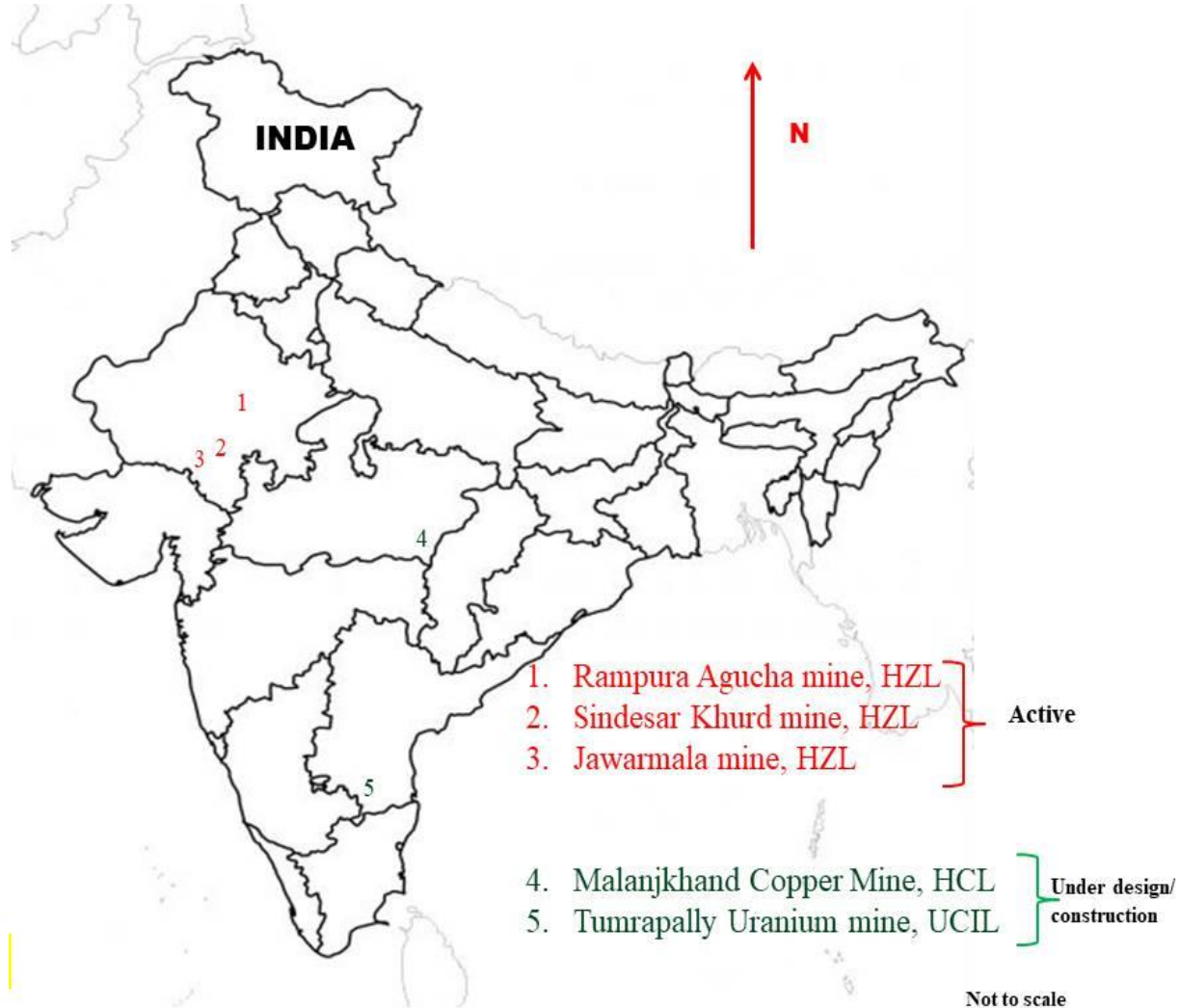
Mine Back filling

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Application of paste backfill: Global scenario



Application of paste backfill: Indian scenario



Historical development of paste backfilling

Mine/Country	Year	Reference
Hardy mine, Falconbridge Nickel Mines Ltd., Sudbury, Canada	1957	Tariq and Yanful (2013)
Bad Grund Mine, Germany	1970s	Tariq and Yanful (2013)
Lucky Friday mine, Idaho, USA	1980s	Brackebusch (1994)
South Africa	1990s	Tariq and Yanful (2013)
Jinchuan Group Co., China	1996	Wu et al. (2017a)
Henty Gold mine, Australia	1997	Henderson et al. (1998)
Brazil	1998	Stone (2014)
Greece	2000s	Stone (2014)
Turkey	2000	Stone (2014)
Tanzania	2001	Qi and Fourie (2019a)
Peru	2005	Qi and Fourie (2019a)
SK mine, HZL, India	2015	Behera et al. (2020a)
Saudi Arabia	2017	Brown et al.(2019)

Application of paste backfill: chronology

